

Aaquib Tabrez

Curriculum Vitae

Postdoctoral Scholar
University of Southern California
✉ tabrez@usc.edu
📄 aaquibtabrez.github.io
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Research Interests

I design autonomous agents and robots that can communicate, coordinate, and teach people, with a focus on making these systems safe, transparent, and trustworthy. I ground my research in interactive tasks involving physical and virtual agents, drawing on methods from human-robot interaction, explainable AI, reinforcement learning, and foundation models.

Professional Experience

2024 – present **University of Southern California**, Postdoctoral Scholar.
Advisor: Prof. Stefanos Nikolaidis

2014 – 2016 **Daimler, Chennai, India**.
Procurement Manager

Education

2019 – 2024 **University of Colorado, Boulder**, GPA: 4.0, Ph.D., Computer Science.
Advisor: Prof. Bradley Hayes
Dissertation: *Mediating Trust and Influence in Human-Robot Teams via Multimodal Communication and Explanation for Mental Model Alignment* ([Link to Dissertation](#))
Committee: Bradley Hayes*, Alessandro Roncone, Daniel Szafir, Nisar Ahmed, & Sonia Chernova

2017–2019 **University of Colorado, Boulder**, GPA: 4.0, MS, Mechanical Engineering.

2010–2014 **National Institute of Technology Karnataka, India**, B.Tech, Mechanical Engineering.

Awards and Recognition

- 2024 **People's Choice Award - 3MT Competition**.
Won the People's Choice Award at the 2024 CU Boulder Three Minute Thesis (3MT) competition, receiving the highest number of audience votes.
- 2023 **Annual Research Expo'23 Poster Presentation Award**.
Received best research poster presentation award at the CU Boulder's Annual Research Expo '23.
- 2022 **Robotics: Science and Systems (RSS) Pioneers**.
Selected for workshop bringing together top early career researchers in robotics.
- 2022 **Best Student Paper Award Runner-up at AAMAS**.
For the paper "Descriptive and Prescriptive Visual Guidance to Improve Shared Situational Awareness in Human-Robot Teaming".
- 2022 **Won Spring 2022 Annual Research Expo Event**.
Received the best poster presentation award at the Spring 2022 Annual Research Expo from CU Boulder.
- 2020 **IBM PhD Fellowship Finalist**.
One of three students nominated by the CS department at CU Boulder.
- 2019 **Best Paper Award Finalist for Technical Advances at ACM/IEEE HRI**.
For the paper "Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning".
- 2019 **Human-Robot Interaction (HRI) Pioneers**.
Selected for workshop bringing together top early career researchers in HRI.

2019 [Awtar and Teji Singh Graduate Fellowship](#).

A \$5,000 fellowship for early career PhD students demonstrating a strong academic and research record.

Papers in Submission

Asterisk (*) denotes shared first authorship

[paper link](#) **Algorithmic Prompt Generation for Diverse Human-like Teaming and Communication with Large Language Models.**

Siddharth Srikanth, Varun Bhatt, Boshen Zhang, Werner Hager, Charles Michael Lewis, Katia P. Sycara, **Aaquib Tabrez**, Stefanos Nikolaidis

[no paper link](#) **Title omitted for blind review.**

Aaquib Tabrez, Ryan Leonard, Bradley Hayes.

Journal Articles

[paper link](#) **A survey of Mental Modeling Techniques in Human-Robot Teaming.**

Aaquib Tabrez, Matthew B. Luebbers, Bradley Hayes.

Springer-Nature Current Robotics Reports, 2020

Conference Publications

[paper link](#) **Recency Bias in Task Performance History Affects Perceptions of Robot Competence and Trustworthiness.**

Matthew B. Luebbers*, **Aaquib Tabrez***, Kanaka Samagna Talanki, Bradley Hayes.

Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2024

[paper link](#) **Autonomous Justification for Enabling Explainable Decision Support in Human-Robot Teaming.**

Aaquib Tabrez*, Matthew B. Luebbers*, Kyler Ruvane*, and Bradley Hayes.

Robotics: Science and Systems (RSS), 2023

[paper link](#) **Descriptive and Prescriptive Visual Guidance to Improve Shared Situational Awareness in Human-Robot Teaming.**

Aaquib Tabrez*, Matthew B. Luebbers*, Bradley Hayes.

International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2022

Best Student Paper Runner-up (Top 2 of 629 submissions).

[paper link](#) **Asking the Right Questions: Facilitating Semantic Constraint Specification for Robot Skill Learning and Repair.**

Aaquib Tabrez*, Jack Kawell*, Bradley Hayes.

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021

[paper link](#) **Explanation-based Reward Coaching to Improve Human Performance via Reinforcement Learning.**

Aaquib Tabrez, Shivendra Agrawal, Bradley Hayes.

ACM/IEEE International Conference on Human Robot Interaction (HRI), 2019

Best Technical Paper Runner-up.

Workshop Publications

[paper link](#) **Hierarchical Multi-Agent Reinforcement Learning with Explainable Decision Support for Human-Robot Teams.**

Aaquib Tabrez*, Matthew B. Luebbers*, Kyler Ruvane*, Ashley H. Rabin, Kevin W. King, William Gerichs, and Bradley Hayes.

Proceedings of the Workshop on Explainability for Human-Robot Collaboration (X-HRI), 2024

[paper link](#) **Autonomous Policy Explanations for Effective Human-Machine Teaming.**

Aaquib Tabrez.

Doctoral Consortium at the AAAI Conference on Artificial Intelligence, 2024

- [paper link](#) **Effective Human-Machine Teaming through Communicative Autonomous Agents that Explain, Coach, and Convince.**
Aaquib Tabrez
Doctoral Consortium at the International Conference on Autonomous Agents and Multiagent Systems, 2023
- [paper link](#) **Augmented Reality and Proxy Grippers Improve Demonstration-based Robot Skill Learning.**
 Carl L. Mueller, Matthew B. Luebbers, **Aaquib Tabrez**, and Bradley Hayes.
Proceedings of the Workshop on Life-Long Learning with Human Help (L3H2), 2023
- [paper link](#) **Mediating Trust and Influence in Human-Robot Interaction via Explainable AI.**
Aaquib Tabrez, Bradley Hayes.
Pioneers Workshop at Robotics: Science and Systems (RSS), 2022
- [paper link](#) **Augmented Reality-Based Explainable AI Strategies for Establishing Appropriate Reliance and Trust in Human-Robot Teaming.**
 Matthew B. Luebbers*, **Aaquib Tabrez***, Bradley Hayes.
Workshop on Virtual, Augmented and Mixed Reality for Human-Robot Interaction (VAM-HRI), 2022
- [paper link](#) **Solutions for Socially Intelligent HRI in Real-World Scenarios (SSIR-HRI).**
 Karen Tatarian, Sera Buyukgoz, Marine Chamoux, **Aaquib Tabrez**, Bradley Hayes, Mohamed Chetouani.
Companion of the ACM/IEEE International Conference on Human-Robot Interaction, 2021
- [paper link](#) **Interactive Constrained Learning from Demonstration Using Visual Robot Behavior Counterfactuals.**
 Carl Mueller, **Aaquib Tabrez**, Bradley Hayes
Workshop on Accessibility of Robot Programming and Work of the Future at RSS, 2021
- [paper link](#) **Emerging Autonomy Solutions for Human and Robotic Deep Space Exploration.**
 Matthew B. Luebbers*, Christine T. Chang*, **Aaquib Tabrez***, Jordan Dixon*, Bradley Hayes.
SpaceCHI: Human-Computer Interaction for Space Exploration, 2021
- [paper link](#) **Automated Failure-Mode Clustering and Labeling for Informed Car-To-Driver Handover in Autonomous Vehicles.**
Aaquib Tabrez*, Matthew B. Luebbers*, Bradley Hayes.
Workshop on Assessing, Explaining, and Conveying Robot Proficiency for Human-Robot Teaming, 2020
- [paper link](#) **Improving human-robot interaction through explainable reinforcement learning.**
Aaquib Tabrez, Bradley Hayes.
Companion of the ACM/IEEE International Conference on Human-Robot Interaction, 2019

Teaching & Research Assistantships

- Spring 2021 - 2024 **Army Research Lab [STRONG Program](#): Strengthening Teamwork for Robust Operations in Novel Groups**, *University of Colorado Boulder, CO.*
Research Assistant, Prof. Bradley Hayes
- Fall 2020 **CSCI 5302/4302: Advanced Robotics**, *University of Colorado Boulder, CO.*
Teaching Assistant, Prof. Bradley Hayes
- Fall 2020 **CSCI 3302: Introduction to Robotics**, *University of Colorado Boulder, CO.*
Teaching Assistant, Prof. Bradley Hayes
- Spring 2020 **CSCI 5922: Neural Networks and Deep Learning**, *University of Colorado Boulder, CO.*
Teaching Assistant, Profs. Adam Bloniarz & Shumin Wu
- Fall 2019 **CSCI 3302: Introduction to Robotics**, *University of Colorado Boulder, CO.*
Teaching Assistant, Prof. Bradley Hayes
- Spring 2019 **CSCI 5322: Algorithmic Human-Robot Interaction**, *University of Colorado Boulder, CO.*
Course Grader, Prof. Bradley Hayes
- Spring 2018 **MCEN-4026: Manufacturing Processes and Systems**, *University of Colorado Boulder, CO.*
Course Grader, Prof. Jenifer Blacklock

Invited Seminars and Colloquia

- 2024 **University of Southern California**, *Multimodal Explanation-based Reward Coaching and Decision Support to Improve Human-Robot Teaming*, Guest Lecture: CSCI 699: Robot Learning.
- 2023 **Talking Robotics**, *Building Trust and Transparency through Explainable Multimodal Communication*.
- 2021 **University of Colorado Boulder**, *Mediating Trust and Influence in Human-Robot Interaction via Explainable AI*, Robotics Summer Seminar Series.

Research Mentorship

- 2024 - **Siddharth Srikanth**, *Undergraduate*, USC.
- 2024 - **Junsoo Kim**, *Undergraduate*, USC.
- 2024 - **Darius Mahjoob**, *Undergraduate*, USC.
- 2024 - **Qingpei Li**, *Masters*, USC.
- 2024 - 2025 **Boshen Zhang**, *Masters*, USC.
- 2024 **David Chaparro**, *Masters*, CU Boulder.
- 2023 - 2024 **Kyler Ruvane**, *Masters*, CU Boulder.
- 2023 **Nathan Howard**, *Masters*, CU Boulder.
- 2022 - 2023 **Kanaka Talanki Sreenivasa Murthy**, *Masters*, CU Boulder.
- 2021 - 2022 **Karthik Siddaramanna**, *Masters*, CU Boulder.
- 2020 - 2021 **Aditi Periyannan**, *Undergraduate*, Tufts University.
- 2019 **Felix Moses**, Berkeley High School.
- 2019 **Stephen Kwak**, Bellarmine High School.
- 2018 - 2019 **Xi Hu**, *Undergraduate*, CU Boulder.

Academic Service, Memberships, and Public Outreach

◦ Workshop & Committee Leadership:

- Co-Organizer, Workshop on Human-Robot Interaction for Explainability in Robotics, *RO-MAN 2023*.
- **Program Committee Chair**, RSS Pioneers 2023 Workshop, *RSS 2023*.
- Co-Organizer, Solutions for socially intelligent HRI in real-world scenarios workshop, *HRI 2021*.
- **Program Chair**, HRI Pioneers 2021 Workshop, *HRI 2021*.
- Co-Organizer, Solutions for socially intelligent HRI in real-world scenarios workshop, *RO-MAN 2020*.
- **Program Chair**, HRI Pioneers 2020 Workshop, *HRI 2020*.

◦ Grant Review: NSF Panelist (2025)

◦ Conference Reviewer: HRI, ICRA, IROS, RO-MAN, CHI, XAIP@ICAPS, XAI@IJCAI

◦ Journal Reviewer: RSS, IEEE RA-L, ACM T-HRI, IJHCI

Media Coverage

- 2024 **Alhub** interviewed me about my research on explainability and human-robot interaction.
- 2024 I was featured by **CU Graduate School** and **CU Boulder Computer Science** after receiving the People's Choice Award for the Three Minute Thesis (3MT) competition.
- 2022 **CU Engineering News** highlighted my research on trust in human-robot interaction using augmented reality-based communication.